

Advanced Programming for HPC

Labwork 2: Get to Know Your GPU

Student: Vu Ha Vy – 2540112

GPU Hardware Summary

The execution output from Google Colab:

Property	Value
Device ID	0
Device Name	Tesla T4
Compute Capability	7.5
PCI Bus ID	0
PCI Device ID	4
UUID	GPU-b48a33b6-9da6-c0ba-7899-678ed7de2bda
Watchdog	Disabled
FP32/FP64 Performance Ratio	32
Multiprocessor Count (SMs)	40
Total CUDA Cores	2560
Free Memory	15,527,116,800 bytes (14.46 GB)
Total Memory	15,637,086,208 bytes (14.56 GB)

Table 1: GPU Overview

```

1  cuda.detect()
2
Found 1 CUDA devices
id 0          Tesla T4          [SUPPORTED]
    Compute Capability: 7.5
    PCI Device ID: 4
    PCI Bus ID: 0
    UUID: GPU-b48a33b6-9da6-c0ba-7899-678ed7de2bda
    Watchdog: Disabled
    FP32/FP64 Performance Ratio: 32
Summary:
1/1 devices are supported
True
3
4
5
6
7  gpu = cuda.select_device(0)
8  gpu.id
9
0
10
11
12  gpu.name
13
'Tesla T4'
14
15
16  sm_count = gpu.MULTIPROCESSOR_COUNT
17  print(sm_count)
18
40
19
20
21  core_count = 64 * sm_count
22  print(core_count)
23
2560
24
25
26  cuda.current_context().get_memory_info()
27
MemoryInfo(free=15527116800, total=15637086208)

```

Figure 1: Captured output of the labwork.